

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 06: Friday 28 August 2026

(Week 3)

Remark. Q05.7 to Q05.14 are leftover exercises from the Tuesday class in Week 3.

Remark. Q05.7 to Q05.9 are on Section 10.3-10.4.

Q05.10 to Q05.14 are on Section 12.1-12.3.



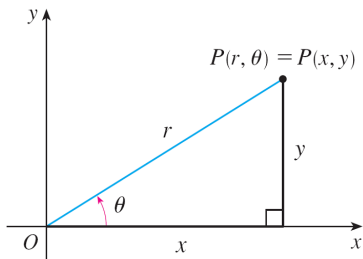
Q06.1 to Q06.6 are on Section 13.1-13.2. More questions on Section 13.1-13.3 will be set for Week 4.

Remark:

If you like to do more exercises, you may

- Try some questions for L1 (Lecture Group 1). Prof Chin and I use different questions for our classes.
- Try some questions in Thomas' Calculus. (Answers to odd-numbered questions are provided at the end of the book.)

Polar coordinates.



Instead of using Cartesian coordinates (x, y) , any point in $\mathbb{R}^2$ can be represented by **polar coordinates** (r, θ) , where

$$r \geq 0$$

- r is the distance between the origin and the point,
- θ is the angle from the positive x -axis to the straight line joining the origin and the point.

Equations relating (x, y) and (r, θ) :

$$x = r \cos \theta, \quad y = r \sin \theta,$$

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}.$$

Q05.7. For the following equations in Cartesian coordinates, rewrite them in terms of polar coordinates:

(i) $\frac{x^2}{9} + \frac{y^2}{4} = 1.$

(ii) $y^2 = 6x.$

Solution: (i)

$$\begin{aligned}\frac{x^2}{9} + \frac{y^2}{4} = 1 &\iff \frac{(r \cos \theta)^2}{9} + \frac{(r \sin \theta)^2}{4} = 1 \\ &\iff 4r^2 \cos^2 \theta + 9r^2 \sin^2 \theta = 36. \\ &\iff 4r^2 + 5r^2 \sin^2 \theta = 36\end{aligned}$$

(ii)

$$y^2 = 6x \quad (1)$$

$$\iff (r \sin \theta)^2 = 6r \cos \theta$$

$$\iff r^2 \sin^2 \theta = 6r \cos \theta.$$

$$\Leftrightarrow r \sin^2 \theta = 6 \cos \theta \quad (2)$$



$$r=0$$

the origin

$$\textcircled{\ast} r=0, \theta=\frac{\pi}{2}$$

is a solution
of the new
equation (2)

$$x(x-1)=0 \quad x^2=x \quad \text{has solns } x=0, x=1$$

$$x=1$$

$$(x, y) = (1, 0) \quad (r, \theta) = (1, 0)$$

Q05.8. A point $\mathbf{P}$ in $\mathbb{R}^2$ is such that both the Cartesian coordinates and polar coordinates of $\mathbf{P}$ are given by the same ordered pair of numbers. Determine the locus of $\mathbf{P}$. First give your answer in terms of polar coordinates, and then give also your answer in terms of Cartesian coordinates.

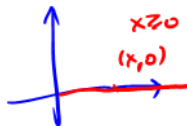
Solution: Let the Cartesian coordinates of $\mathbf{P}$ be (x, y) , and the polar coordinates of $\mathbf{P}$ be (r, θ) . Then $\mathbf{P}$ satisfies

$$(x, y) = (r, \theta) \iff \begin{cases} x = r, & (1) \\ y = \theta. & (2) \end{cases}$$

Handwritten notes:

- From (1): $r \cos \theta = r$
- From (2): $r \sin \theta = \theta$
- From (1): $r(1 - \cos \theta) = 0$
- From (2): $r = 0, \theta = 0$

$$(x, y) = (r, \theta) \iff \begin{cases} x = r, & (1) \\ y = \theta. & (2) \end{cases}$$



From (1), we have

$$x = r \implies \cancel{x^2} = r^2 = \cancel{x^2} + y^2 \implies y^2 = 0 \implies y = 0.$$

Substituting $y = 0$ into (2), we get

$$\theta = 0,$$

which corresponds to all the points on the positive x -axis (including the origin).

Now observe that all these points satisfy (1) and (2).

Thus the locus of P is given by polar equation $\theta = 0$.

Then the locus of **P** in Cartesian coordinates is given by

$$y = 0 \text{ and } x \geq 0.$$

Q05.9. (a) For each part below, sketch the curve defined by the given polar equation (i.e. equation in polar coordinates):

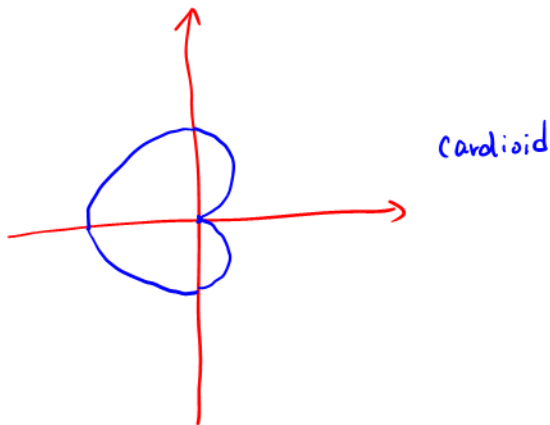
(i) $r = 2(1 - \cos \theta)$;

(ii) $r = 6 \sin \theta$.

(b) For both parts (i) and (ii), rewrite the equation in Cartesian coordinates.

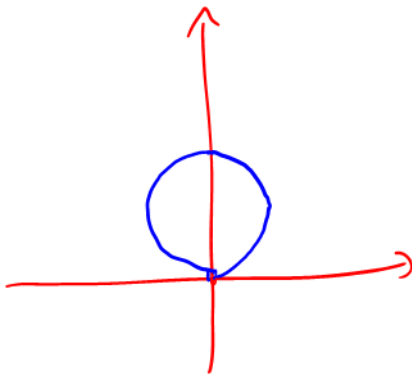
(a) (i) $r = 2(1 - \cos \theta)$

Solution:



(a) (ii) $r = 6 \sin \theta$.

Solution:



(b) For both parts (i) and (ii), rewrite the equation in Cartesian coordinates.

(i) $r = 2(1 - \cos \theta)$ (1)

$$x = r \cos \theta$$

$$\Leftrightarrow r^2 = 2(r - r \cos \theta)$$

$$\Leftrightarrow x^2 + y^2 = 2(\sqrt{x^2 + y^2} - x) \quad (2)$$

Is $r=0$ a solution
of (1)?

Yes, $(r, \theta) = (0, 0)$
is a solution of (1).

$$(ii) \ r = 6 \sin \theta. \quad (1)$$

$$\Leftrightarrow r^2 = 6r \sin \theta$$

$$\Leftrightarrow x^2 + y^2 = 6y \quad (2)$$

$$\Leftrightarrow x^2 + y^2 - 6y + 9 = 9$$

$$\Leftrightarrow x^2 + (y-3)^2 = 3^2$$

circle centered at $(0, 3)$ and of radius 3.

$$y = r \sin \theta$$

Is $r=0$ a solution of (1)?

Yes, $(r, \theta) = (0, 0)$

is a solution of (1).

Q05.10 to Q05.14 are on Section 12.1 -12.3.

Vector-valued functions of one variable

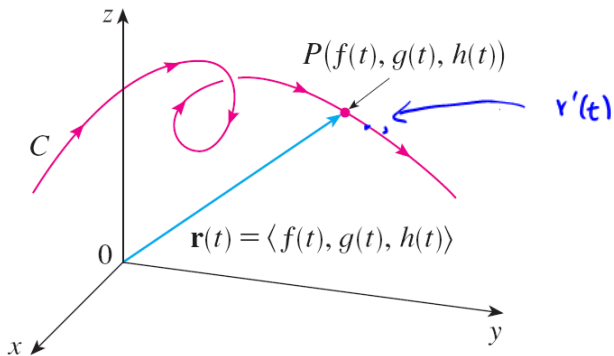
A vector-valued function of one variable is of the form

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

or

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$$

for some scalar function f , g and h (called the component functions of $\mathbf{r}$).



As $\mathbf{t}$ varies, $\mathbf{r}(\mathbf{t})$ traces out a curve $\mathbf{C}$ (in $\mathbb{R}^2$ or $\mathbb{R}^3$), and we say that $\mathbf{r}(\mathbf{t})$ is a parametrization of $\mathbf{C}$.

Given a vector function

$$\mathbf{v}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}.$$

Its derivative, indefinite integral and definite integral are defined by using the corresponding objects for each component function:

$$\mathbf{v}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k};$$

$$\int \mathbf{v}(t) dt = \left(\int f(t) dt \right) \mathbf{i} + \left(\int g(t) dt \right) \mathbf{j} + \left(\int h(t) dt \right) \mathbf{k};$$

$$\int_a^b \mathbf{v}(t) dt = \left(\int_a^b f(t) dt \right) \mathbf{i} + \left(\int_a^b g(t) dt \right) \mathbf{j} + \left(\int_a^b h(t) dt \right) \mathbf{k}.$$

Remark. For indefinite integral, each component indefinite integral has its own integration constant.

Q05.10. Consider the vector function

$$\mathbf{v}(t) = (3t^2 + 2t)\mathbf{i} + \pi \sin(\pi t)\mathbf{j} + e^t\mathbf{k}.$$

(i) Evaluate the derivative $\mathbf{v}'(t)$. Find also $\mathbf{v}'(2)$.

(ii) Evaluate the indefinite integral $\int \mathbf{v}(t) dt$.

(iii) Evaluate the definite integral $\int_0^1 \mathbf{v}(t) dt$.

Consider the vector function

$$\mathbf{v}(t) = (3t^2 + 2t)\mathbf{i} + \pi \sin(\pi t)\mathbf{j} + e^t\mathbf{k}.$$

(i) Evaluate the derivative $\mathbf{v}'(t)$. Find also $\mathbf{v}'(2)$.

Solution:

$$\mathbf{v}'(t) = (6t + 2)\mathbf{i} + \pi^2 \cos(\pi t)\mathbf{j} + e^t\mathbf{k}.$$

$$\begin{aligned}\mathbf{v}'(2) &= (6 \cdot 2 + 2)\mathbf{i} + \pi^2 \cos(2\pi)\mathbf{j} + e^2\mathbf{k} \\ &= 14\mathbf{i} + \pi^2\mathbf{j} + e^2\mathbf{k}.\end{aligned}$$

$$(ii) \mathbf{v}(t) = (3t^2 + 2t)\mathbf{i} + \pi \sin(\pi t)\mathbf{j} + e^t\mathbf{k}.$$

$$\begin{aligned} & \int \mathbf{v}(t) dt \\ &= \left(\int (3t^2 + 2t) dt \right) \mathbf{i} + \left(\int \pi \sin(\pi t) dt \right) \mathbf{j} + \left(\int e^t dt \right) \mathbf{k}; \\ &= (t^3 + t^2 + C_1) \mathbf{i} + (-\cos(\pi t) + C_2) \mathbf{j} + (e^t + C_3) \mathbf{k}. \end{aligned}$$

$$\begin{aligned} (iii) & \int_0^1 \mathbf{v}(t) dt \\ &= \left(\int_0^1 (3t^2 + 2t) dt \right) \mathbf{i} + \left(\int_0^1 \pi \sin(\pi t) dt \right) \mathbf{j} + \left(\int_0^1 e^t dt \right) \mathbf{k} \\ &= \left[t^3 + t^2 \right]_0^1 \mathbf{i} + \left[-\cos(\pi t) \right]_0^1 \mathbf{j} + \left[e^t \right]_0^1 \mathbf{k}; \\ &= ((1 + 1) - 0) \mathbf{i} + (-\cos \pi + \cos 0) \mathbf{j} + (e - e^0) \mathbf{k} \\ &= 2\mathbf{i} + 2\mathbf{j} + (e - 1)\mathbf{k}. \end{aligned}$$

Q05.11. Consider the curve in $\mathbb{R}^3$ given by

$$\mathbf{r}(t) = (\ln t)\mathbf{i} + \frac{t-1}{t+3}\mathbf{j} + (t \ln t)\mathbf{k}.$$

Find parametric equations of the tangent line to the curve at the point where $t = 1$.

Q05.11. $\mathbf{r}(t) = (\ln t)\mathbf{i} + \frac{t-1}{t+3}\mathbf{j} + (t \ln t)\mathbf{k}.$

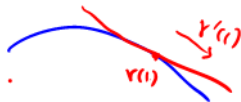
Solution: First

$$\begin{aligned}\mathbf{r}'(t) &= \frac{1}{t}\mathbf{i} + \frac{1 \cdot (t+3) - (t-1) \cdot 1}{(t+3)^2}\mathbf{j} + (1 \cdot \ln t + t \cdot \frac{1}{t})\mathbf{k} \\ &= \frac{1}{t}\mathbf{i} + \frac{4}{(t+3)^2}\mathbf{j} + ((\ln t) + 1)\mathbf{k}\end{aligned}$$

Now

$$\mathbf{r}(1) = (\ln 1)\mathbf{i} + \frac{1-1}{1+3}\mathbf{j} + (1 \ln 1)\mathbf{k} = 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}.$$

$$\mathbf{r}'(1) = \frac{1}{1}\mathbf{i} + \frac{4}{(1+3)^2}\mathbf{j} + ((\ln 1) + 1)\mathbf{k} = \mathbf{i} + \frac{1}{4}\mathbf{j} + \mathbf{k}.$$



$$\mathbf{r}(1) = 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}, \quad \mathbf{r}'(1) = \mathbf{i} + \frac{1}{4}\mathbf{j} + \mathbf{k}.$$

Thus parametric equations of the tangent line to the curve at the point where $t = 1$ are given by

$$x = 0 + 1 \cdot t, \quad y = 0 + \frac{1}{4}t, \quad z = 0 + 1 \cdot t$$

$$\iff x = t, \quad y = \frac{1}{4}t, \quad z = t.$$

Q05.12. A particle moves in $\mathbb{R}^3$ such that its velocity at time t is given by

$$\mathbf{v}(t) = (6t^2 + 4t)\mathbf{i} + (8t^3 + 2t)\mathbf{j} + \frac{1}{1+t^2}\mathbf{k}.$$

If the position of the particle at time $t = 0$ is $\mathbf{i} + 2\mathbf{j} + \pi\mathbf{k}$. Find the position of the particle at time $t = 1$.

[Remark: The position $\mathbf{r}(t)$ and velocity $\mathbf{v}(t)$ (as vector functions of time t) of the particle satisfy $\mathbf{r}'(t) = \mathbf{v}(t)$.]

Supplementary Exercise to Q05.12: Find also the speed and acceleration of the particle at time $t = 1$.

$$\mathbf{v}(t) = (6t^2 + 4t)\mathbf{i} + (8t^3 + 2t)\mathbf{j} + \frac{1}{1+t^2}\mathbf{k}.$$

$$\mathbf{r}'(t) = \mathbf{v}(t), \quad \mathbf{r}(0) = \mathbf{i} + 2\mathbf{j} + \pi\mathbf{k}.$$

Thus

$$\begin{aligned} \mathbf{r}(t) &= \int \mathbf{v}(t) dt \\ &= \int ((6t^2 + 4t)\mathbf{i} + (8t^3 + 2t)\mathbf{j} + \frac{1}{1+t^2}\mathbf{k}) dt \\ &= (2t^3 + 2t^2 + C_1)\mathbf{i} + (2t^4 + t^2 + C_2)\mathbf{j} \\ &\quad + (\tan^{-1}(t) + C_3)\mathbf{k} \end{aligned}$$

Then

$$\begin{aligned} \mathbf{r}(0) &= (0 + 0 + C_1)\mathbf{i} + (0 + 0 + C_2)\mathbf{j} + (\tan^{-1}(0) + C_3)\mathbf{k} \\ \Rightarrow \mathbf{i} + 2\mathbf{j} + \pi\mathbf{k} &= C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k} \\ \Rightarrow C_1 &= 1, \quad C_2 = 2, \quad C_3 = \pi. \end{aligned}$$

Thus

$$\mathbf{r}(t) = (2t^3 + 2t^2 + 1)\mathbf{i} + (2t^4 + t^2 + 2)\mathbf{j} + (\tan^{-1}(t) + \pi)\mathbf{k}.$$

So

$$\begin{aligned}\mathbf{r}(1) &= (2 + 2 + 1)\mathbf{i} + (2 + 1 + 2)\mathbf{j} + (\tan^{-1}(1) + \pi)\mathbf{k} \\ &= 5\mathbf{i} + 5\mathbf{j} + \frac{5\pi}{4}\mathbf{k}.\end{aligned}$$

$\frac{5\pi}{4}$

$$\mathbf{v}(t) = (6t^2 + 4t)\mathbf{i} + (8t^3 + 2t)\mathbf{j} + \frac{1}{1+t^2}\mathbf{k}.$$

Let $\mathbf{s}(t)$ and $\mathbf{a}(t)$ be the speed and acceleration of the particle at time t respectively. Then $\mathbf{a}(t) = \mathbf{v}'(t)$

$$\mathbf{s}(t) = \|\mathbf{v}(t)\| = \sqrt{(6t^2 + 4t)^2 + (8t^3 + 2t)^2 + \frac{1}{(1+t^2)^2}},$$

$$\begin{aligned}\mathbf{a}(t) = \mathbf{v}'(t) &= \frac{d}{dt}((6t^2 + 4t)\mathbf{i} + (8t^3 + 2t)\mathbf{j} + \frac{1}{1+t^2}\mathbf{k}) \\ &= (12t + 4)\mathbf{i} + (24t^2 + 2)\mathbf{j} - \frac{2t}{(1+t^2)^2}\mathbf{k}.\end{aligned}$$

$$\text{Then } \mathbf{s}(1) = \sqrt{100 + 100 + \frac{1}{4}} \approx 14.15,$$

$$\mathbf{a}(1) = 16\mathbf{i} + 26\mathbf{j} - \frac{1}{2}\mathbf{k}.$$

Q05.13. Two particles move in $\mathbb{R}^3$ such that their positions at time t are given by:

$$\mathbf{r}_1(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k},$$

$$\mathbf{r}_2(t) = (1 + 2t)\mathbf{i} + (1 + 6t)\mathbf{j} + (1 + 14t)\mathbf{k}.$$

- (a) Do the particles collide? Justify your answer.
- (b) Do their paths intersect? Justify your answer.

$$\mathbf{r}_1(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k},$$

$$\mathbf{r}_2(t) = (1 + 2t)\mathbf{i} + (1 + 6t)\mathbf{j} + (1 + 14t)\mathbf{k}.$$

Solution: (a) For particles to collide, we equate the two vector functions using the *same* parameter t :

$$\mathbf{r}_1(t) = \mathbf{r}_2(t).$$

Equating the three components, we get

$$\left\{ \begin{array}{ll} t = 1 + 2t, & (1) \\ t^2 = 1 + 6t, & (2) \\ t^3 = 1 + 14t. & (3) \end{array} \right. \Rightarrow t = -1$$

This system does not have solution. To see this, from (1), we get $t = -1$. But $t = -1$ does not satisfy (2), since $(-1)^2 = 1 \neq 1 + 6(-1) = -5$.

(Similarly $t = -1$ does not satisfy (3).)

So, we conclude that the 2 particles do not collide.

$$\mathbf{r}_1(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k},$$

$$\mathbf{r}_2(t) = (1 + 2t)\mathbf{i} + (1 + 6t)\mathbf{j} + (1 + 14t)\mathbf{k}.$$

$$\mathbf{r}_2(s) = (1 + 2s)\mathbf{i} + (1 + 6s)\mathbf{j} + (1 + 14s)\mathbf{k}$$

(b) For the paths to intersect, we equate the two vector functions using *different* parameters $\mathbf{s}$ and t :

$$\mathbf{r}_1(t) = \mathbf{r}_2(\mathbf{s}).$$

Equating the components, we get

$$\left\{ \begin{array}{ll} t = 1 + 2s, & (1) \end{array} \right.$$

$$\left\{ \begin{array}{ll} t^2 = 1 + 6s, & (2) \end{array} \right.$$

$$\left\{ \begin{array}{ll} t^3 = 1 + 14s. & (3) \end{array} \right.$$

This system has two solutions given by

$$\mathbf{s} = 0, t = 1 \quad \text{and} \quad \mathbf{s} = \frac{1}{2}, t = 2.$$

$$t = 1 + 2s$$

To see this, we substitute (1) into (2), and we get

$$(1 + 2s)^2 = 1 + 6s \implies 4s^2 - 2s = 0$$

$$\implies 2s(2s - 1) = 0 \implies s = 0, \frac{1}{2}.$$

Substituting $s = 0$ into (1), we get $t = 1$.

Substituting $s = \frac{1}{2}$ into (1), we get $t = 1 + 2 \cdot \frac{1}{2} = 2$.

Thus we get two potential solutions given by

$$s = 0, t = 1 \quad \text{and} \quad s = \frac{1}{2}, t = 2.$$

Now we substitute $\mathbf{s} = \mathbf{0}$, $\mathbf{t} = \mathbf{1}$ into (1), (2), (3) to see that $\mathbf{s} = \mathbf{0}$, $\mathbf{t} = \mathbf{1}$ is indeed a solution of the system of the three equations. (This amounts to checking that

$$1 = 1 + 2 \cdot 0, \quad 1^2 = 1 + 6 \cdot 0, \quad 1^3 = 1 + 14 \cdot 0.)$$

Similarly, we substitute $\mathbf{s} = \frac{1}{2}$, $\mathbf{t} = \mathbf{2}$ into (1), (2), (3) to see that $\mathbf{s} = \frac{1}{2}$, $\mathbf{t} = \mathbf{2}$ is indeed a solution of the system of the three equations. (This amounts to checking that

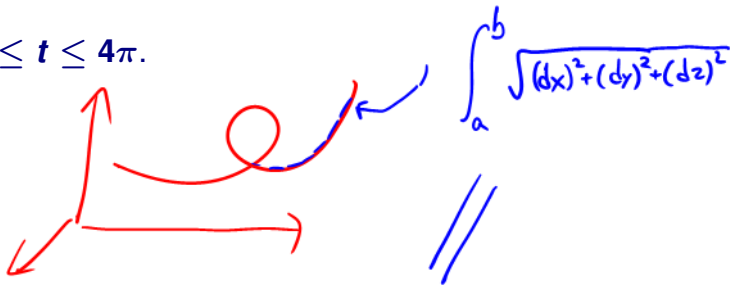
$$\overset{\checkmark}{2} = 1 + 2 \cdot \frac{1}{2}, \quad \overset{\checkmark}{2^2} = 1 + 6 \cdot \frac{1}{2}, \quad \overset{\checkmark}{2^3} = 1 + 14 \cdot \frac{1}{2}.)$$

In summary, the two paths intersect at the two points $(1, 1, 1)$ and $(2, 4, 8)$.

Q05.14. Find the arc length of the curve

$$x = 2t - 2 \sin t, \quad y = 2 - \sqrt{2} \cos t, \quad z = \sqrt{2} \cos t$$

for $0 \leq t \leq 4\pi$.



$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Solution: From the equation of given curve, we have

$$x = 2t - 2 \sin t \implies \frac{dx}{dt} = 2 - 2 \cos t,$$

$$y = 2 - \sqrt{2} \cos t \implies \frac{dy}{dt} = \sqrt{2} \sin t,$$

$$z = \sqrt{2} \cos t \implies \frac{dz}{dt} = -\sqrt{2} \sin t,$$

Thus the arc length of the curve is given by

$$\begin{aligned} L &= \int_0^{4\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \\ &= \int_0^{4\pi} \sqrt{(2 - 2 \cos t)^2 + (\sqrt{2} \sin t)^2 + (-\sqrt{2} \sin t)^2} dt \\ &= \int_0^{4\pi} \sqrt{4 - 8 \cos t + 4 \cos^2 t + 2 \sin^2 t + 2 \sin^2 t} dt \\ &= \int_0^{4\pi} \sqrt{8 - 8 \cos t} dt. \end{aligned}$$

Using the identity $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$ (with $\theta = \frac{t}{2}$), we have

$$\begin{aligned} L &= \int_0^{4\pi} \sqrt{8(1 - \cos t)} \, dt \\ &= 4 \int_0^{4\pi} \sqrt{\sin^2\left(\frac{t}{2}\right)} \, dt = 4 \int_0^{4\pi} \left| \sin\left(\frac{t}{2}\right) \right| \, dt \end{aligned}$$

Note that for $0 \leq t \leq 2\pi$, we have $\sin\left(\frac{t}{2}\right) \geq 0$; and for $2\pi \leq t \leq 4\pi$, we have $\sin\left(\frac{t}{2}\right) \leq 0$. Therefore,

$$\begin{aligned} \int_0^{4\pi} \left| \sin\left(\frac{t}{2}\right) \right| \, dt &= \int_0^{2\pi} \sin\left(\frac{t}{2}\right) \, dt + \int_{2\pi}^{4\pi} -\sin\left(\frac{t}{2}\right) \, dt \\ &= \left[-2 \cos\left(\frac{t}{2}\right) \right]_0^{2\pi} + \left[2 \cos\left(\frac{t}{2}\right) \right]_{2\pi}^{4\pi} \\ &= 2 - (-2) + 2 - (-2) = 8. \end{aligned}$$

Hence the arc length of the curve is $L = 4 \cdot 8 = 32$.

Q06.1 to **Q06.6** are on Section 13.1-13.2.

Functions of ~~several~~ variables, and their limits

Several



2 or 3

Q06.1. Consider the function $f(x, y, z) = \frac{x - y}{y^2 + z^2}$.

Evaluate $f(3, -1, 2)$ and $f(0, -\frac{1}{3}, 0)$.

Solution:

$$f(3, -1, 2) = \frac{3 - (-1)}{(-1)^2 + 2^2} = \frac{4}{5};$$

$$f(0, -\frac{1}{3}, 0) = \frac{0 - (-\frac{1}{3})}{(-\frac{1}{3})^2 + 0^2} = 3.$$

Q06.2. Consider the function $f(x, y) = \frac{1}{\ln(9 - x^2 - y^2)}$.

Find and sketch the maximal domain of the function.

Solution: Note that $\ln t$ is defined iff $t > 0$, and $\frac{1}{r}$ is defined iff $r \neq 0$. Thus $f(x, y)$ is defined if and only if

$$9 - x^2 - y^2 > 0 \text{ and } \ln(9 - x^2 - y^2) \neq 0$$

$$\iff x^2 + y^2 < 9 \text{ and } 9 - x^2 - y^2 \neq 1$$

$$\iff \underline{x^2 + y^2 < 9 \text{ and } x^2 + y^2 \neq 8.}$$



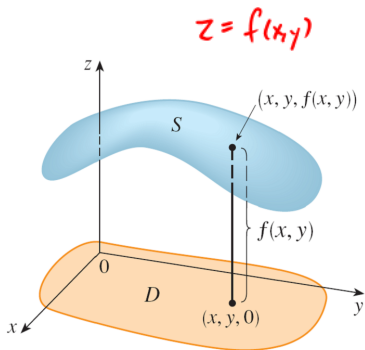
$$y = f(x)$$

One way of visualizing $f(\mathbf{x}, \mathbf{y})$ is to draw its graph.

If $f(\mathbf{x}, \mathbf{y})$ is a function of two variables with domain $D \subseteq \mathbb{R}^2$, then the **graph** of f is the set of all points $(\mathbf{x}, \mathbf{y}, \mathbf{z}) \in \mathbb{R}^3$ such that $\mathbf{z} = f(\mathbf{x}, \mathbf{y})$ and $(\mathbf{x}, \mathbf{y}) \in D$.

Typically the graph of a function f of two variables is a **surface S** in $\mathbb{R}^3$ defined by the equation $\mathbf{z} = f(\mathbf{x}, \mathbf{y})$.

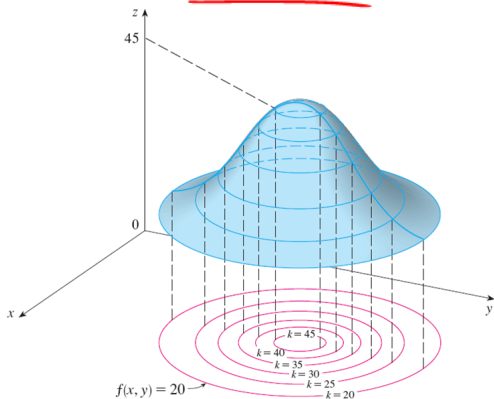
We can visualize the graph **S** of f as lying directly above or below its domain D in the xy -plane.



Graphing functions of two variables is not simple!

Nonetheless, we do gain some insight into f by examining its level curves.

A level curve of $f(x, y)$ is the curve in the xy -plane defined by the equation $f(x, y) = k$ for some constant k .



Q06.3. Consider the function $f(x, y) = \frac{x + y}{x - y}$.

Sketch the level curves $f(x, y) = k$ for $k = -2, 0, 1, 2$.

Solution: Note that $f(x, y) = \frac{x + y}{x - y}$ is defined only when $x - y \neq 0$ (i.e. $x \neq y$).

For $k \in \mathbb{R}$ and $x \neq y$, we have

$$\begin{aligned} f(x, y) &= k \\ \iff \frac{x + y}{x - y} &= k \\ \iff x + y &= kx - ky \\ \iff (1 + k)y &= (k - 1)x \\ \iff y &= \frac{(k - 1)x}{1 + k} \quad \text{for } k \neq -1. \end{aligned}$$

Thus for $k = -2$, the level curve is $y = 3x$ with $y \neq x$ (i.e. with $(x, y) \neq (0, 0)$).

For $k = 0$, the level curve is $y = -x$ with $y \neq x$ (i.e. with $(x, y) \neq (0, 0)$).

For $k = 1$, the level curve is $y = 0$ with $y \neq x$ (i.e. with $(x, y) \neq (0, 0)$).

For $k = 2$, the level curve is $y = \frac{x}{3}$ with $y \neq x$ (i.e. with $(x, y) \neq (0, 0)$).

